addac112 NULLSTATION

This is the addac112 looper firmware with a different engine in the left knob section: **RADIO**, a port of the Nullstation plugin (`../PLUGINSto112/Nullstation 3`).

RADIO is a radio that never finds a station. It generates everything that lives between stations, and runs it through a simulated receiver:

- hiss, static crashes, hum and fading;
- heterodyne whistles, voices in no language, distant music;
- Morse, data modes, radar, whistlers, jammers and interference.

Your input, or the selected loop, can be one of the stations on the dial.

It was built from FRIPPER (`addac112_fripper`), with the TAPE engine replaced by RADIO. The looper, banks, SD card, presets, loop pitch, sync and REC prob/delay are the same as in the stock firmware.

```
engine/     radio_engine.* (DSP), generators.h, bands.h, primitives.h, radio_param_map.h
(knobs, settings, preset keys);
            no firmware dependencies
reference/  the plugin's engine and tests, unchanged (the A/B reference)
test/       desktop tests of the engine: ./run_tests.sh; Cortex-M7 cycle estimate: m7/
firmware/   the 112 firmware (src/, web/ editor, Makefile)
```

```
cd firmware
make                # build/NULLSTATION_<version>.bin (git hash, or the build date)
make program-app    # bootloader in DFU mode
```

It builds against addac112's libdaisy / DaisySP / CMSIS (`make ADDAC112_DIR=...` to override).

Switching engines

Each engine is its own firmware. This one always runs RADIO. Plain granular is the stock addac112 firmware.

1. Put the firmware `.bin` files (112, CD SKIP, FRIPPER, NULLSTATION, ...) in a folder named `ENGINES` on the SD card.
2. Go to **Menu** → **ENGINE** (the last item of the main menu). It lists the `.bin` files in `ENGINES`.
3. Choose one and confirm **FLASH & RESTART? YES**.

Presets store both engines' values: the stock grain keys plus `radio_*` keys. The stock 112 firmware ignores the `radio_*` keys, but if it saves a preset, those keys are dropped from that preset.

RADIO controls (dual-label overlay)

Row	Col 1	Switch (FREE / ÷8 / ÷4)	Col 2	Col 3
1	POSITION / **DIAL**	START GRID / **MODE** (AM / SSB / FM)	POS DEV / **DRIFT**	GRAINS ACTIVE / **BAND** (jack in trigger mode: **SEEK**)
2	PLAY LENGTH / **BANDWIDTH**	SIZE GRID / **SPEAKER** (DIRECT / TABLETOP / WALKIE)	LENGTH DEV / **FADING**	ATTACK / **ENV**
3	GRAIN DELAY / **ACTIVITY**	SKIP CLOCK / **EVENT SYNC** (FREE / ÷8 / ÷4)	DELAY DEV / **DENSITY**	DECAY / **TONE**
4	DIRECTION / **SCAN**	REPEAT / **MAP** (FIXED / WANDER)	PROB/REPEAT / **SQUELCH**	GRAIN PITCH / **FINE** (CV: 1 V = 1 kHz)

Row	Col 1	Switch (FREE / ÷8 / ÷4)	Col 2	Col 3
5	VOLUME MIN / **INPUT**	–	VOLUME DEV / **HISS**	PANNING / **WIDTH**

Knobs

- **DIAL**: tuning across the band.
  - The pot has a little hysteresis (0.06 % of the band), so its noise doesn't wobble the tuning. FINE tunes in between.
  - GLIDE (in the settings) smooths the moves. Turning past signals makes them whistle by (SCAN SWEEPS sets how loud).
- **BAND**: 8 steps: VLF, LW, MW, SW, CB, FM, AIR, UHF. Each band has its own signals, noise colour and occurrences.
  - VLF: whistlers and dawn chorus.
  - LW: beacons.
  - MW: far AM stations.
  - SW: everything.
  - CB: skip voices.
  - FM: capture and popcorn.
  - AIR: transmissions behind the squelch.
  - UHF: digital buzz.

  The jack, summed with the knob, selects bands. With **SEEK INPUT** on, the jack is a trigger: each rising edge seeks a signal and stays on it until DIAL is turned.
- **DRIFT**: the first half is WANDER (an unstable oscillator). The second half adds DRIFT (an elastic random scan around the set frequency, up to 1/16 of the band).
- **BANDWIDTH**: 200 Hz to 16 kHz, with 3 kHz at the centre. It sets how much of the dial you hear at once.
- **FADING**: signals fade in and out, and a notch sweeps through the sound.
- **ENV**: scales SWELL TIME, HOLD and RELEASE of every occurrence, ×0.1 to ×10 (×1 at the centre).
- **ACTIVITY**: occurrences per minute, from 1.8 to 98 (off at the minimum).
- **DENSITY**: how crowded the band is.
- **TONE**: a lowpass after the speaker, 1.5 kHz to open.
- **SCAN**:
  - Right of the centre, the receiver sweeps the band back and forth.
  - Left of the centre, it seeks, hopping from signal to signal (every 7 s down to 0.6 s).
  - The centre (±8 %) is off.
- **SQUELCH**: off, then S0 to S9+. Below the threshold it is silent; only strong signals and transmissions open it.
- **FINE**: ±5 kHz around DIAL. The GRAIN PITCH CV adds 1 kHz per volt.
  - In SSB this shifts the pitch of what you hear, so an LFO on it gives a warbling SSB voice.
  - With QUANTIZE on (QUANTIZER: BOTH or RADIO ONLY), the tuning snaps to the band's channel raster.
- **INPUT**: how strong your audio is as a station: –inf to +6 dB, with 0 dB at about 80 % of the knob. See *Live input*.
- **HISS**: the noise floor (–inf to +6 dB).
- **WIDTH**: the stereo spread of the signals, noise and occurrences.

Switches

- **MODE**:
  - AM: a lone carrier only quiets the noise; two carriers beat into a whistle.
  - SSB: signals are frequency-shifted by their offset + BFO, the classic Donald Duck sound.
  - FM: the strongest signal captures the rest, and weak signals crackle.

The MODE SWITCH setting turns the centre position into CW (700 Hz filter).

- **SPEAKER**: DIRECT / TABLETOP / WALKIE-TALKIE, or POCKET / CLOCK RADIO / HEADSET (SPEAKERS setting).
- **EVENT SYNC**:
  - With a loop playing, occurrences and seek hops wait for the next ÷8 or ÷4 slice of the selected loop.
  - The Poisson timing still decides *whether* one happens; the slice decides *when*.
- **MAP**:
  - FIXED: every frequency always has the same signals (LANDSCAPE number).
  - WANDER: the band reinvents itself around the dial. Signals appear on the side you are turning towards, live for a while and fade, with the occurrence envelope.

Other panel controls in RADIO mode

- **GRAINS VOLUME** is the radio level.
- **GRAINS FEEDBACK** records the radio into the loop.
- **GRAINS trigger out** sends a pulse when an occurrence starts and on every seek landing.
- **Encoder + STOP** (special function, enabled in Options) goes to the next LANDSCAPE. In WANDER it gives a new set of signals.

Settings

- Two quick pages, scrolled from the main screen:
  - **RADIO SETTINGS**: SEEK INPUT, SOURCE, CARRIER, FOLLOW DIAL, INPUT AT, MODE SWITCH, SPEAKERS, LANDSCAPE.
  - **RADIO SETTINGS 2**: DRIVE, AGC, CRACKLE, HUM, FADE RATE, BFO, GLIDE, MAP DRIFT.
- Every setting, by group, under **Menu → Options → 5.RADIO OPTIONS**:
  - **SIGNALS**: CARRIERS, VOICES, MUSIC, DATA, MORSE, EXOTIC, INTERFERENCE. These are volumes, and also how many of each kind live on the band (OFF: none).
  - **OCCURRENCES**: CRASHES, GLITCHES, DROPOUTS, SWELLS, SWEEPS, TRANSMISSIONS, BURSTS. Louder types happen more often; OFF never.
  - **ENVELOPE**: SWELL TIME, HOLD, RELEASE, RANDOMIZE.
  - **RECEIVER**: MODE SWITCH (SSB / CW), BFO, AGC, DRIVE, GLIDE, SPEAKERS.
  - **ATMOSPHERE**: CRACKLE, HUM, BUZZ, MAINS 50/60, FADE RATE, FLUTTER, POPS, SCAN SWEEPS, MAP DRIFT.
  - **INPUT**: SEEK INPUT, SOURCE, FOLLOW DIAL, INPUT AT, CARRIER.
  - **LANDSCAPE**: 1–99.

Display (RADIO)

- **Main screen:**
  - A header line with the band and frequency, what you are hearing (e.g. `RTTY +350`), and `SQ` while the squelch is closed.
  - The band scope below it:
    - a tick per signal (height = strength; dotted for broadband ones);
    - the needle (double) where the receiver really is;
    - a dotted line where the dial is set;
    - the DRIFT range on the top edge;
    - a small marker where the input sits when FOLLOW DIAL is off.
- **Left bar:** the S-meter, with a line at the SQUELCH threshold. It blinks while the squelch is closed.
- **Right bar:** WIDTH.
- **Top bar:** the band in two letters (VL LW MW SW CB FM AR UH), or `SK` in seek mode (inverted while the input is high).
- **Bottom bar:**
  - the MODE (inverted in WANDER);

    - the occurrence envelope (swell / hold / release);
    - F = FINE (knob + CV) in 100 Hz steps.
- **Readout:** turning an overlay knob or flipping a switch shows its name and value for 1.5 s, with `PICKUP` until the knob reaches a preset's value.

Live input

The input is a station on the dial. VOL DRY stays clean; what the radio does to your sound is in GRAINS VOLUME.

- **SOURCE**: what INPUT puts on the dial: the LIVE input, the LOOP as you hear it, or BOTH.
- **FOLLOW DIAL** on (default): your audio is always tuned in. FOLLOW DIAL off: it sits at **INPUT AT**, and tuning matters:
    - **AM**: it fades out with the BANDWIDTH filter as you tune away.
    - **SSB / CW**: it is frequency-shifted by (dial − input) + BFO. DIAL and FINE become a frequency shifter on your playing, and FINE's CV can play it.
    - **FM**: off-tune gives distortion and noise.
    - It whistles past while the dial moves over it.
- It gets the same treatment as every signal:
    - its own fading and flutter;
    - dropouts;
    - the receiver filter, AGC (it pumps the band) and the fading notch;
    - squelch;
    - hum, glitches (stutter, crush, chop, warble of your playing), drive, speaker, tone and width.
- **CARRIER: KEYED** (default) works like keying a transmitter:
    - The input's carrier follows how loud you play (–50 to –20 dBFS, 5 ms attack, 80 ms release).
    - While you play, the hiss quiets (strongly in FM) and the S-meter rises.
    - With SQUELCH up the radio stays silent until you play, then opens with your sound. It closes about 0.3 s after you stop.
    - **STEADY** is the plugin's behaviour: as long as INPUT is up, the carrier is there.
- **In FM** the input competes with the stations. INPUT at 0 dB is as strong as a strong station, so at +6 dB your playing captures most of the band, and a weak input is swallowed by a strong station.
- **GRAINS FEEDBACK** prints the radio, with your processed input, into the loop. Re-record it through the radio again and it degrades a little more each pass.
- There is no added latency. It is sample by sample, and only the glitch stutter reads from the past.

Sound against the plugin

The engine is the plugin's DSP, with:

- fixed arrays instead of the heap;
- fast sine, tanh and pow on the audio path;
- the 112 additions, each of which can be switched off (`Params::exactAU`).

`test/ab_test` renders 42 cases on both engines with the same seed: every band × mode, scan, seek, drift, occurrences, squelch, hum, input, Wander, wide VLF.

- With the plugin's arithmetic (`–DRADIO_FAST_MATH=0`) the output is identical, sample for sample. Wander differs only by rounding, from a double-precision clock.
- With the fast math the 112 runs, the level is the same to 0.1 dB and the zero-crossing rate (brightness) is the same. The difference is –50 to –125 dB below the signal; the worst cases are SW and Wander, where rounding flips a random decision.

The plugin's own tests (every band and mode, occurrence envelopes and rates, FM clicks, scan sweeps and squelch, drift) pass on the 112 engine. `test/features_test` covers the 112 additions, and `test/rate_test` covers every sample rate of the 112.

CPU

The radio files are built with `-O2` (the rest of the firmware, like the stock 112, with `-Og`).

These are estimates from running the Cortex-M7 build in an emulator (`test/m7/m7_cycles.py`): executed instructions, with VDIV/VSQRT weighted at 14 cycles. Read them as ±30 %.

Case (48 kHz)	M7 load
MW, AM, defaults	~21 %
SW, SSB, defaults	~34 %
VLF, 16 kHz bandwidth (wide whistlers)	~40 %
SW, 16 kHz, occurrences at max, input parked	~46 %
SW, 16 kHz, dense band, fast sweep (12 signals)	~52 %
the same, the CPU guard at 4 signals	~31 %
FRIPPER TAPE, 4 reversed heads, -O2 (-Og in its firmware: ~53 %)	*~39 %*

- Each signal being rendered costs about 250 cycles per sample, on top of a base of about 2100.
- **CPU guard**: above 85 % CPU the bridge lowers the number of signals rendered (the loudest ones stay), from 12 down to 4, and above 4 it allows only 4 occurrences at once. Below 65 % it raises them again, slowly.
- At **96 kHz** the radio runs at 48 kHz inside, with 47-tap half-band filters in and out (flat to 20 kHz, –63 dB above 28 kHz). That keeps the same CPU share as at 48 kHz (~52 % in the worst case above).
- At 8–44.1 kHz it runs at that rate.

Memory

	GRANULAR (stock)	RADIO
Loop memory	360 s	~371 s (mono at 44.1 kHz)
Engine buffers (1.5 MB: spot generators, occurrences, glitch buffers)	–	top of the loop array in SDRAM
Per-spot state the engine reads every 0.7 ms + bridge (~40 KB)	–	heap (RAM_D2), created at boot
Grain voice reserve (21 KB heap)	yes	no

- DTCM is at 89.1 % (stock: 88.6 %).
- The binary is 449 KB of the 480 KB the bootloader copies to SRAM, which leaves ~41 KB for more code.

Web editor

`firmware/web/` has two Radio sections (knobs and settings) next to Grains, with the levels in dB. The ENGINE row of the card tab is gone, because nothing reads `engine_mode` any more.

Run the editor's tests with `for t in firmware/web/test/*.test.js; do node "$t"; done`. The sample files are read from `../addac112/support`. The tests check two things:

- the editor's `radio_*` keys are exactly the ones `radio::saveKeys()` writes;
- its defaults are the engine's (`test/run_tests.sh`).

Not done yet

- Not yet run on hardware. Things to check:
  - CPU % in the worst cases above, and the guard;
  - heap headroom with large banks;
  - the DIAL hysteresis against the real pot noise;
  - the FINE CV scaling;

    - the SEEK input;
    - the trigger out;
    - EVENT SYNC with the loop;
    - the band scope at 30 fps.
- Overlay artwork: needs the panel dimensions and knob positions.